

Assignment: Analysis of Functional Connectivity from Resting-state fMRI

Overview / Learning objectives

Students will learn to:

1. Obtain ADNI / Other Dataset of resting fMRI data and prepare it in BIDS format.
2. Preprocess rs-fMRI using a reproducible pipeline (fMRIPrep).
3. Perform quality control (motion, tSNR, outlier volumes).
4. Compute functional connectivity (seed-based, ROI-ROI correlation, whole-brain connectivity matrices).
5. Perform basic group-level statistics and visualize results.
6. Interpret common pitfalls (motion confound, smoothing choice, multiple comparisons).

Prerequisites / Software

- Linux or macOS (or WSL2 on Windows)
- Docker or Singularity (recommended) or conda
- fMRIPrep (Docker/Singularity)
- Nilearn (Python), Nibabel, numpy, pandas, matplotlib, scikit-learn, networkx
- Optional: AFNI/FSL/FreeSurfer (for some QC or registration steps)
- ADNI account (registration & data use agreement required) to download data / Other Datasets

Dataset

ADNI resting-state fMRI — students should:

1. Register at the ADNI portal and request access to resting fMRI series for a small subset (e.g., 10 subjects: 5 controls, 5 patients).
2. Download the raw DICOMs or NIfTI. Convert to BIDS (recommended: dcm2niix + BIDS layout).

Important note: ADNI data require account and data use agreement — make sure you have permission before downloading.

Assignment tasks (step-by-step)

Task 0 — Prepare BIDS dataset

- Convert DICOM → NIfTI (dcm2niix).
- Organize BIDS folder structure:

```
/myproject/bids/  
  sub-01/  
    func/  
      sub-01_task-rest_bold.nii.gz  
      sub-01_task-rest_bold.json  
    anat/
```

```
sub-01_T1w.nii.gz
sub-01_T1w.json
sub-02/...
dataset_description.json
participants.tsv
```

- Add `participants.tsv` with group labels (control/patient) and relevant covariates (age, sex).

Deliverable: BIDS dataset folder with at least 8–12 subjects.

Task 1 — Preprocessing with fMRIPrep (recommended)

Use fMRIPrep to do reproducible preprocessing (slice timing, motion correction, spatial normalization, coregistration, and confound estimates).

Docker example command (replace paths):

```
docker run --rm -it \
-v /path/to/bids:/data:ro \
-v /path/to/derivatives:/out \
-v /path/to/freesurfer:/fs \
poldracklab/fmriprep:latest \
/data /out participant \
--fs-license-file /fs/license.txt \
--nthreads 8 \
--omp-nthreads 8 \
--ignore slicetiming \
--output-spaces MNI152NLin2009cAsym:res-2 \
--use-aroma
```

Notes:

- `--use-aroma` runs ICA-AROMA for data-driven denoising.
- Provide FreeSurfer license if doing surface outputs.
- If slice timing differs or unknown, adjust `--ignore slicetiming` accordingly.

Outputs you will use:

- preprocessed BOLD in MNI (`..._desc-preproc_bold.nii.gz`)
 - confound regressors TSV files (`..._desc-confounds_regressors.tsv`)
 - brain masks, T1w skull-stripped images, reports (HTML)
-

Task 2 — Quality Control (QC)

For each subject:

1. Inspect fMRIPrep HTML report (registration, fieldmaps, motion).
2. Compute summary motion metrics from confounds file:
 - Framewise displacement (FD) mean and max.
 - Number of volumes with $FD > 0.5$ mm (or other threshold).
3. Compute temporal SNR (tSNR): $\text{mean}(\text{signal}) / \text{sd}(\text{signal})$ across time (within brain mask).
4. Consider excluding subjects with excessive motion (e.g., $>20\%$ volumes $FD > 0.5\text{mm}$) or flag them.

Deliverable: QC table (CSV/TSV) with subject, mean_FD, max_FD, n_fd_gt_0.5, tSNR, excluded_flag.

Task 3 — Nuisance regression & temporal filtering

Choose one of two denoising approaches (explain your choice in report):

A. ICA-AROMA + minimal confounds

- Use fMRIPrep `--use-aroma` outputs (denoised BOLD) and additionally regress out high motion volumes (scrubbing) if needed and apply bandpass.

B. Confound regression (aCompCor + motion)

- From `desc-confounds_regressors.tsv` select:
 - 6 motion parameters + their derivatives
 - aCompCor components (e.g., first 5 CSF+WM comps)
 - Global signal (optional; mention controversy)
- Regress out chosen confounds using linear regression and then bandpass filter (e.g., 0.01–0.08 Hz).

Python (conceptual) snippet for confound regression + filtering:

(Full code later in Task 4.)

Task 4 — Compute Functional Connectivity

Perform **three** FC analyses:

A. Seed-based correlation

- Choose seed(s): e.g., posterior cingulate cortex (PCC) for default mode network.
- Extract seed time series (mean within a sphere or atlas ROI).
- Compute Pearson correlation with every voxel → Fisher Z transform → group maps.

B. ROI-ROI connectivity matrix

- Use an atlas (e.g., AAL, Harvard-Oxford, or Schaefer 200-parcel).
- Extract mean time series per ROI; compute pairwise Pearson correlations; Fisher Z transform.
- Produce connectivity matrices and summary measures (e.g., mean within-network connectivity).

C. Data-driven: ICA or network identification

- Run spatial ICA (e.g., Nilearn `CanICA` or `DictLearning`) to get components; identify components matching known networks (DMN, sensorimotor).
 - Compute dual regression or component time series correlations.
-

Task 5 — Group statistics / hypothesis testing

Propose a simple hypothesis:

- Example: "Group A (controls) will have higher DMN intra-network connectivity than Group B (patients)."

Steps:

1. For seed maps: run a voxelwise two-sample t-test (e.g., using `nilearn.mass_univariate.permuted_OLS` or `fsl_randomise`).

2. For ROI-ROI: for each edge (or summary metric), run group t-test with multiple comparisons correction (FDR or Bonferroni) or permutation testing.
3. Adjust for covariates (age, sex, mean FD) in GLM.

Multiple comparisons: Use FDR correction ($q < 0.05$) or cluster-based correction for voxelwise tests.

Task 6 — Graph theoretical analysis (optional / bonus)

- Construct adjacency matrices using thresholded connectivity (e.g., proportional threshold 10–20%).
 - Compute graph metrics: degree, clustering coefficient, global efficiency, modularity (use `networkx` or `bctpy`).
 - Compare graph measures between groups.
-

Example code snippets

These are compact examples — adapt paths and variable names to your dataset.

1) Extract ROI time series and compute ROI-ROI connectivity (Nilearn)

```
# run in a Jupyter cell
import numpy as np
import pandas as pd
from nilearn import image, input_data, plotting
from scipy.stats import zscore
import matplotlib.pyplot as plt

# paths (example)
bold_img = '/out/sub-01/func/sub-01_task-rest_space-MNI152NLin2009cAsym_desc-preproc_bold.nii.gz'
mask_img = '/out/sub-01/func/sub-01_task-rest_space-MNI152NLin2009cAsym_desc-brain_mask.nii.gz'
atlas_img = '/path/to/atlas.nii.gz' # e.g. MNI atlas labels
labels = pd.read_csv('/path/to/atlas_labels.csv') # id -> name

# Extract mean time series per ROI
masker = input_data.NiftiLabelsMasker(labels_img=atlas_img, standardize=True,
                                       detrend=True, low_pass=0.08,
                                       high_pass=0.01, t_r=2.0)
time_series = masker.fit_transform(bold_img)

# compute correlation matrix
corr = np.corrcoef(time_series.T)
z_corr = np.arctanh(corr) # Fisher z

# save
np.save('sub-01_roi_corr.npy', z_corr)

# quick plot
plt.imshow(z_corr, vmin=-1, vmax=1)
plt.title('ROI-ROI connectivity (Fisher Z)')
plt.colorbar()
plt.show()
```

2) Seed-based map (Nilearn)

```
from nilearn.input_data import NiftiSpheresMasker
from nilearn import plotting

seed_coords = [(0, -52, 26)] # PCC example in MNI coords
seed_masker = NiftiSpheresMasker(seed_coords, radius=6, standardize=True,
                                  detrend=True, low_pass=0.08, high_pass=0.01,
                                  t_r=2.0)
```

```

seed_ts = seed_masker.fit_transform(bold_img)

# whole brain masker (use brain mask)
brain_masker = input_data.NiftiMasker(mask_img=mask_img, standardize=True,
                                       detrend=True, low_pass=0.08,
                                       high_pass=0.01, t_r=2.0)
brain_ts = brain_masker.fit_transform(bold_img)

# compute correlation map
import numpy as np
corr_map = np.dot(brain_ts.T, seed_ts)[:,:0] # Pearson assuming standardized
# convert to image
corr_img = brain_masker.inverse_transform(corr_map)
zmap = np.arctanh(corr_img.get_fdata()) # fisher z
z_img = image.new_img_like(corr_img, zmap)

plotting.plot_stat_map(z_img, title='Seed (PCC) correlation',
display_mode='z', cut_coords=6)
plotting.show()

```

3) Group stats example (ROI-based) — permutation test (scipy/sklearn)

```

from scipy.stats import ttest_ind
from sklearn.utils import resample

# assume subjects x edges array: data_control, data_patient (each shape
n_sub, n_edges)
# do two-sample t-test per edge
tvals, pvals = ttest_ind(data_control, data_patient, axis=0)

# FDR correction (Benjamini-Hochberg)
from statsmodels.stats.multitest import multipletests
rej, pvals_fdr, _, _ = multipletests(pvals, alpha=0.05, method='fdr_bh')

```

Quality checks & recommended thresholds

- Mean FD per subject: report and consider excluding >0.2 mm (or at minimum flag subjects >0.5 mm).
 - Volumes with FD > 0.5 mm flagged for scrubbing.
 - tSNR: compute and report distribution; extremely low tSNR indicates acquisition issues.
 - Visual check: registration of BOLD→T1 and T1→MNI should be accurate.
-

Deliverables (what students must submit)

1. **BIDS dataset** (or a manifest + script showing how you organized data).
 2. **Preprocessing script / command** (the exact fMRIPrep command and options used).
 3. **QC table** (CSV/TSV): subject, group, mean_FD, max_FD, n_fd_gt_0.5, tSNR, excluded_flag.
 4. **Functional connectivity outputs:**
 - Seed maps (NIfTI images) and group contrasts (NIfTI).
 - ROI-ROI connectivity matrices (per subject) and group summary matrices.
 - Plots: connectivity matrices, seed correlation map(s), distributions of mean connectivity per network.
 5. **Statistical analysis:** code + results (tables of significant edges or clusters, corrected p-values).
 6. **Short report (2–4 pages):** methods, preprocessing choices & justification, results, limitations, and conclusions.
 7. (Bonus) Graph metrics and group comparisons.
-

Grading rubric (suggested)

- Data preparation and BIDS correctness (10 pts)
 - Preprocessing pipeline & reproducibility (20 pts)
 - QC reporting & justification for exclusions (15 pts)
 - FC computations (seed & ROI) correctness (20 pts)
 - Statistical analysis & correction for multiple comparisons (15 pts)
 - Plots, visualization, and report clarity (15 pts)
 - Bonus: graph analysis & extra exploration (up to +5 pts)
-